

IT 462 – BIG DATA SYSTEMS

Bike-Sharing Demand Analysis



Section 72889 – Group 2

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Date

May 11, 2026



Problem & Motivation



Problem Statement

Predict hourly bike demand to improve bike redistribution and service availability.

Why It Matters



Dynamic demand across stations



Better bike redistribution



Improved urban mobility planning

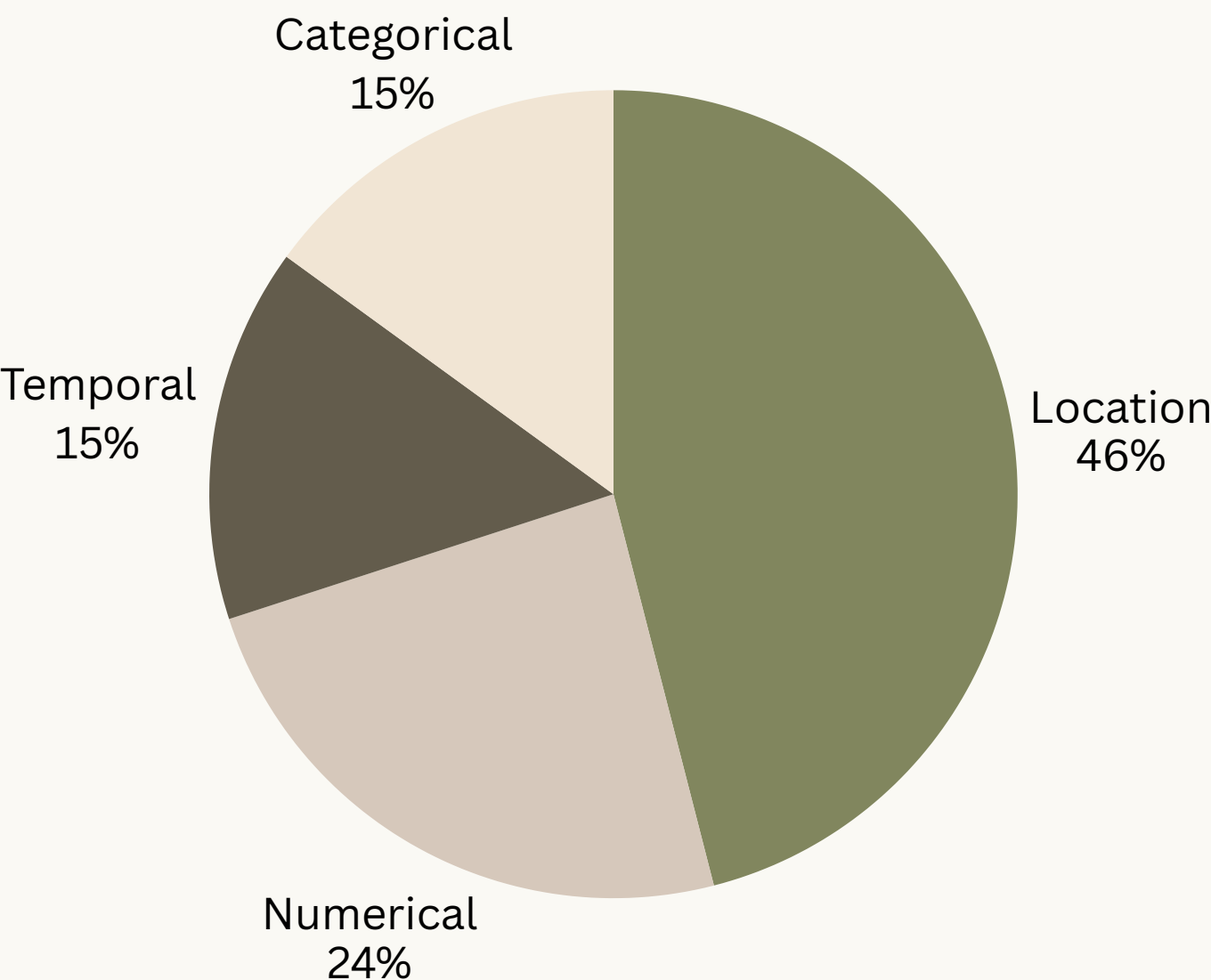
Stakeholders

- Bike-sharing operators

- Transportation planners

Dataset Overview

Distribution of Dataset Feature Types



Source

Citi Bike System Data

Domain

**Urban Mobility & Bike-Sharing
Trips in New York City**

Size

- Rows: **10,240,572**
- Columns: **13**
- Time Span: **Oct 2025 – Dec 2025**

Main Data Quality Issues

- 28,295 **missing station-name** values
- 531 **invalid trip durations** (>24h or negative)
- 597 **boundary-crossing** records outside the target period

Key Preprocessing Steps

- **Removed null and invalid** records
- Applied **temporal filtering** (Oct–Dec 2025)
- Applied 30% **stratified sampling** (seed = 42)

Data Quality & Preprocessing

Phase	Original rows	After cleaning	After temporal filtering	After sampling	After aggregation
Count	10,240,572	10,212,686	10,212,089	3,065,487	2,989,829

RDD Exploration

- 01 **map**
extract hourly demand values
- 02 **reduceByKey**
aggregate total trips by hour
- 03 **sortBy**
identify peak demand hours

```
val hourlyDemandSorted = baseRDD
  .map { record =>
    val totalTrips = record.classicTrips + record.electricTrips
    (record.hour, totalTrips)
  }
  .reduceByKey(_ + _)
  .sortBy({ case (_, totalTrips) => totalTrips }, ascending = false)

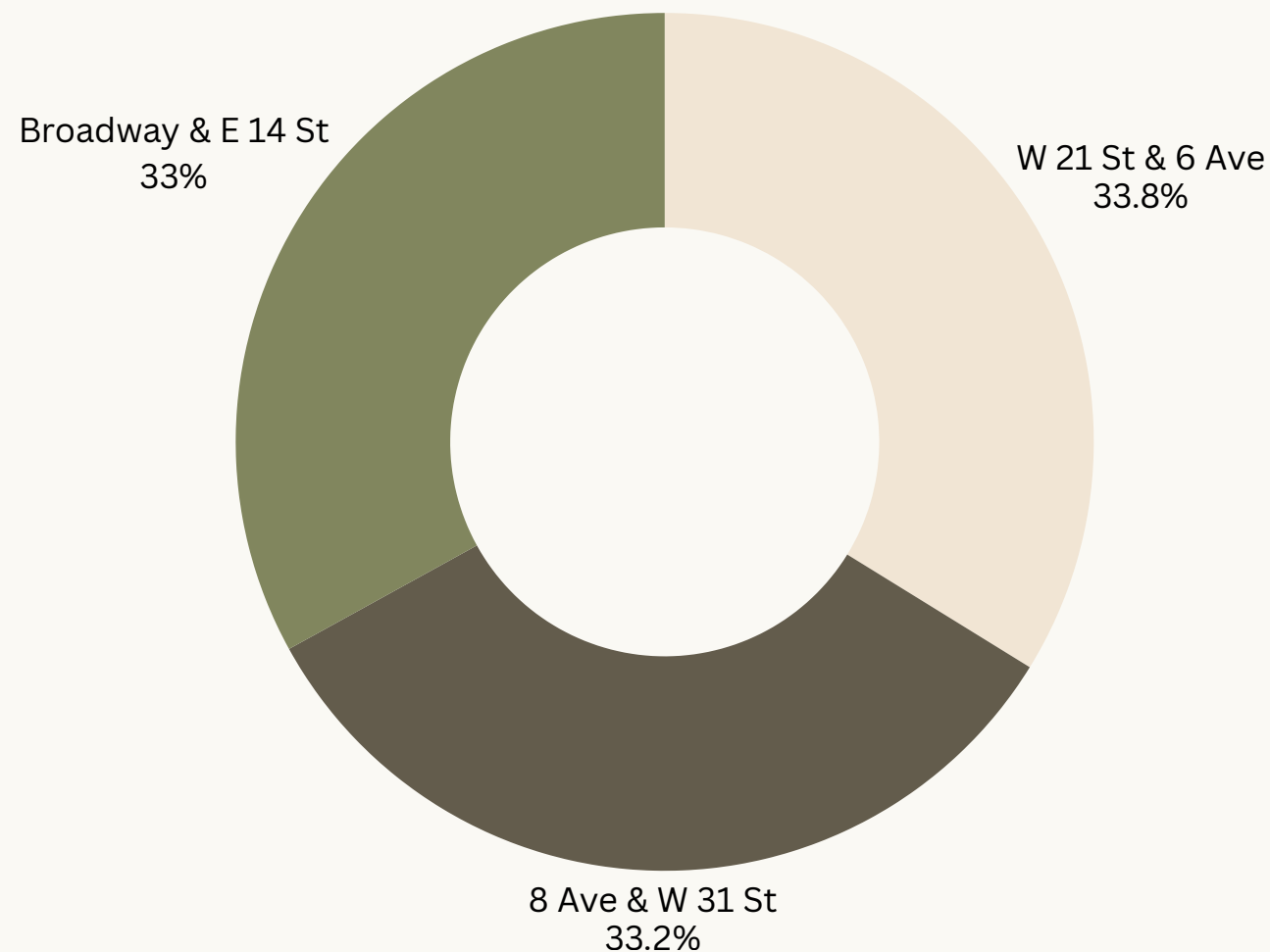
val top10Hours = hourlyDemandSorted.take(10)
```

Peak demand occurred at 17:00 with 276,627 total trips, followed by 18:00 and 16:00.



SQL Analytics Highlights

Electric Bike Share by Station



Key SQL Queries

A

Station Imbalance Analysis

Identify stations with the highest outgoing vs incoming trip imbalance.

B

Bike-Type Preference Analysis

Analyze electric-bike demand at high-traffic stations.

C

Demand Stability Analysis

Compare stable vs peak demand hours across the week.

Interpretation

Electric bikes were consistently preferred at major stations, while rush-hour demand showed higher variability across the week.

ML APPROACH

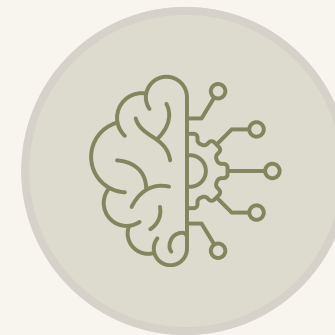


PROBLEM TYPE



Regression — predicting hourly bike demand.

MODELS USED



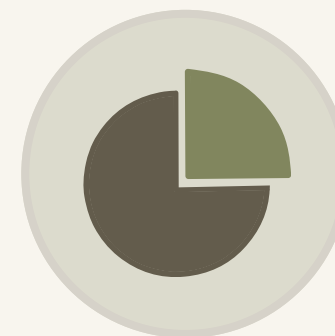
- Linear Regression
- Random Forest Regressor
- Gradient Boosted Trees (GBT)

KEY FEATURES



- Station location
- Hour of day
- Day of week
- Rider type
- Bike type

TRAIN/TEST SPLIT



70% training — 30% testing
Random seed = 42

RESULTS & MODEL PERFORMANCE

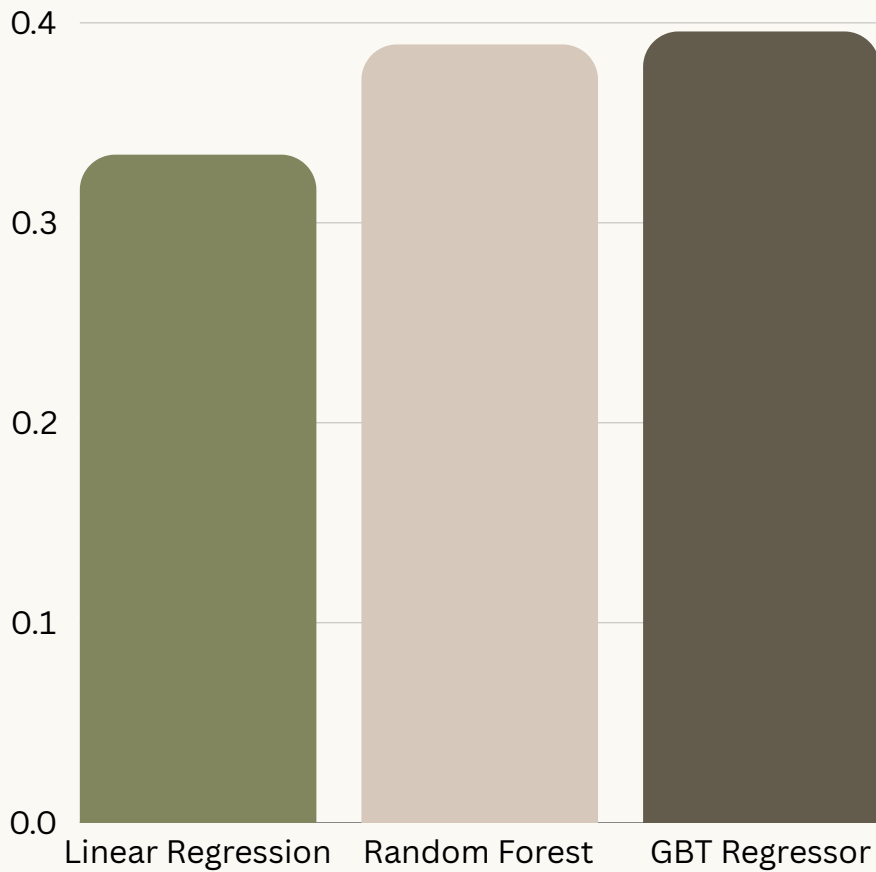


METRIC TABLE

Model	RMSE	MAE	R ²
Linear Regression	6.8473	3.1574	0.3341
Random Forest	6.5552	2.8861	0.3892
GBT Regressor	6.5208	2.8014	0.3957



MODEL COMPARISON (R² SCORE)



INTERPRETATION

GBT achieved the best prediction performance, while Random Forest also outperformed the baseline Linear Regression model.

Interpretation & Insights

01

Key Features

- Station location was the strongest predictor.
 - Demand changed significantly by hour.
-

02

Model Performance

- The model captured regular demand patterns well.
 - Performance dropped on rare peak-demand cases.
-

03

Practical Implications

- Improve bike redistribution during peak hours.
- Support better station balancing decisions.

Limitations & Future Work

A.



B.

Limitations

- The dataset did not include weather or event data.
- Peak-demand cases were difficult to predict accurately.
- No hyperparameter tuning was applied.

Future Work

- Incorporate weather and holiday features.
- Explore advanced models such as LSTM.
- Implement real-time demand prediction.



CONCLUSION



Large-scale analysis

We processed and analyzed over 10 million Citi Bike trip records using Apache Spark.



Key insights

Peak demand occurs during evening hours, and station location is the strongest predictor of bike demand.



Predictive performance

Gradient Boosted Trees (GBT) achieved the best performance for hourly bike demand prediction.



KEY TAKEAWAY

Data-driven demand prediction can support smarter bike redistribution and improve urban mobility planning.



Questions?

